

ECON 3350/7350 Applied Econometrics for Macroeconomics and Finance

Lecture 6: Cointegrating Relations

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Cointegration

Recommended readings

Author	Title	Chapter	Call No
Enders	Applied Econometric Time Series, 4e	6.1, 6.2	HB139 .E55 2015
Verbeek	A Guide to Modern Econometrics	9.2, 9.3	HB139 .V465 2012

Spurious Regression

What are the implications of working with $I(1)$ variables?

In general, **linear combinations** of $I(1)$ variables yields another $I(1)$ variable; however, in some cases the linear combination can result in $I(0)$.

Spurious regression: conclude there is a significant relationship between $I(1)$ variables when in reality there isn't one.

Cointegration: $I(1)$ variables are related such that there exists a linear combination that yields a $I(0)$ process.

Spurious Regression

Let $x_t = x_{t-1} + \varepsilon_{x,t}$ and $y_t = y_{t-1} + \varepsilon_{y,t}$, where $\varepsilon_{x,t}$ and $\varepsilon_{y,t}$ are independent white noise.

Since x_t and y_t are independent random walks, there is no relationship between them.

However, regressing $y_t = \beta_0 + \beta_1 x_t + \varepsilon_t$ typically yields large, significant $\hat{\beta}_1$.

The sampling distribution of $\frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)}$ under $H_0 : \beta_1 = 0$ is not Student-t; it is more dispersed so that

$$\Pr \left(\left| \frac{\hat{\beta}_1}{se(\hat{\beta}_1)} \right| > t_{\alpha, T-k} \right) = \Pr(\text{reject } H_0 \mid H_0 \text{ is true}) > \alpha.$$

So, a *t*-test will tend to reject H_0 incorrectly (i.e. α is not the correct significance level)..

Spurious Regression

In fact, as $T \rightarrow \infty$, the quantity $\sqrt{T}(\hat{\beta}_1 - \beta_1) \rightarrow \infty$.

- t -statistic of β_1 increases with sample size—more data exacerbates the problems with t -tests.
- OLS is not consistent, and there is no well-defined sampling distribution.

Significant t -tests occur because even pure random walks tend to resemble trends; when two $I(1)$ variables **wander in the same direction** for a while, we might mistake this for **cointegration**.

But in $y_t = \beta_0 + \beta_1 x_t + \varepsilon_t$, the residual ε_t is $I(1)$, so would expect low Durbin-Watson statistics.

Practical signs of a **spurious regression**:

- Significant t -statistics; high R^2 values; low Durbin-Watson statistics.

Cointegration

Consider the following example:

$$y_t = \xi_t + \nu_{y,t},$$

$$x_t = \frac{1}{\beta} \xi_t + \nu_{x,t},$$

$$\xi_t = \xi_{t-1} + \eta_t,$$

where the errors $\nu_{y,t}$, $\nu_{x,t}$ and η_t are given by

$$\nu_{y,t} = -\beta \varepsilon_{1,t}, \quad \nu_{x,t} = -\frac{1}{\beta} \varepsilon_{2,t}, \quad \eta_t = \beta \varepsilon_{1,t} + \varepsilon_{2,t},$$

and $\varepsilon_{1,t}$, $\varepsilon_{2,t}$ are both (uncorrelated) white noise processes.

Observe that

$$\begin{aligned}\Delta y_t &= \eta_t + \nu_{y,t} - \nu_{y,t-1}, \\ &= \varepsilon_{2,t} + \beta \varepsilon_{1,t-1} \equiv u_{y,t}, \\ \Delta x_t &= \eta_t + \nu_{x,t} - \nu_{x,t-1}, \\ &= \varepsilon_{1,t} + \frac{1}{\beta} \varepsilon_{2,t-1} \equiv u_{x,t}.\end{aligned}$$

When examined individually, $y_t = y_{t-1} + u_{y,t}$ and $x_t = x_{t-1} + u_{x,t}$ are random walks.

But the linear combination

$$y_t - \beta x_t = \nu_{y,t} - \beta \nu_{x,t} = -\beta \varepsilon_{1,t} + \varepsilon_{2,t}$$

is also I(0).

Cointegration and Common Stochastic Trends

In this example, y_t and x_t are **cointegrated**.

- While each process is a RW, they in fact wander together.
- There exists a **long-run** stable relationship between y_t and x_t , such that they never wander too far away from each other.

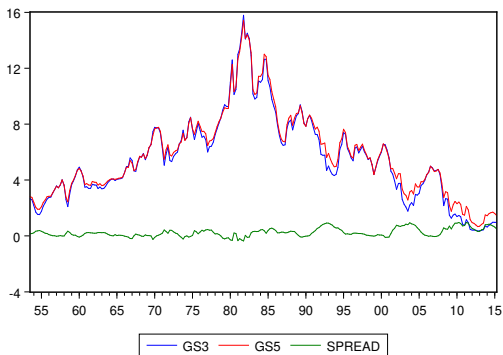
The key component that ties together y_t and x_t is the **common stochastic trend** ξ_t .

- Both y_t and x_t actually embody stationary movements around the non-stationary ξ_t .

In economics, many theories suggest that observed economic data are driven by common stochastic trends—variables evolve together over time and there exist long-run equilibria.

Common Stochastic Trend Example

Expectation Theory of the Term Structure of Interest Rates suggests rates are $I(1)$, but spreads are $I(0)$.



- GS3: ADF statistic is -2.178 (p -value 0.4995); GS5: ADF statistic is -1.977 (p -value 0.6104)
- SPREAD: ADF statistic is -5.608 (p -value 0.0000)

Definition of Cointegration

Let $w_t = (w_{1,t}, \dots, w_{n,t})'$ be a $n \times 1$ vector with variables $w_{i,t}$ integrated of order d .

Definition.

The components of w_t are **cointegrated** of order d, b , denoted $CI(d, b)$, iff

1. all components of w_t are $I(d)$, and
2. there exists a $n \times 1$ vector $\beta \neq 0$ such that

$$z_t = \beta' w_t \sim I(d - b), \quad b > 0.$$

The vector β is called the **cointegrating vector**.

Cointegration Rank

For any given $n \times 1$ vector w_t , there may exist $0 \leq r \leq n - 1$ **linearly independent** cointegrating vectors $\beta_1, \dots, \beta_r$. We say that w_t has **cointegration rank** r .

Suppose $w_{i,t} \sim I(1)$ for $i = 1, \dots, n$ and let $\mathbf{B} = (\beta_1, \dots, \beta_r)$ be a $n \times r$ matrix of cointegrating vectors. Then

$$\mathbf{z}_t = \mathbf{B}' w_t$$

is a $r \times 1$ vector with elements $z_{j,t} \sim I(0)$.

If w_t has **cointegration rank** r , then there are exactly $n - r$ **common stochastic trends** driving the variables in w_t .

Real Business Cycles Example

King, R. G., C. I. Plosser, J. H. Stock, and M. W. Watson (1991). "Stochastic trends and economic fluctuations." *The American Economic Review*, 81, 819-840.

General class of RBC models can be characterized by

$$Y_t = \lambda_t K_t^\theta N_t^{1-\theta},$$
$$\ln \lambda_t = \mu_\lambda + \ln \lambda_{t-1} + \xi_t,$$

where Y_t is output, K_t is capital stock, N_t is labor input, and $\ln \lambda_t$ is log of **total factor productivity**, which is a RW with drift.

The **resource constraint** implies $Y_t = C_t + I_t$, where C_t is consumption and I_t is investment.

Real Business Cycles Example

$\ln \lambda_t / \theta$ is a **common stochastic trend** with growth rate $(\mu_\lambda + \xi_t) / \theta$.

Slight extension of the *Neoclassical Growth Model* yields that the **great ratios** $\frac{C_t}{Y_t}$ and $\frac{I_t}{Y_t}$ are **stationary stochastic processes**.

Therefore, theory implies that $\ln C_t$, $\ln I_t$, and $\ln Y_t$ are all $I(1)$ but the quantities $\ln C_t - \ln Y_t$ and $\ln I_t - \ln Y_t$ are $I(0)$.

That is, there exist two cointegrating vectors such that

$$\mathbf{B}' w_t = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} \ln C_t \\ \ln I_t \\ \ln Y_t \end{pmatrix} = \begin{pmatrix} \ln C_t - \ln Y_t \\ \ln I_t - \ln Y_t \end{pmatrix} \sim I(0).$$

KPSW use this to compute **impulse responses** of economic variables to **productivity shocks**, identified as innovations in the **common stochastic trend**. Such IRFs are features of the **business cycle**.

Testing for Cointegration

When $I(1)$ variables cointegrate, a linear combination will be $I(0)$; otherwise all linear combinations will be $I(1)$.

Basic idea to test for cointegration: given a vector w_t , estimate the regression

$$w_{n,t} = \beta_0 + \beta_1 w_{1,t} + \cdots + \beta_{n-1} w_{n-1,t} + \varepsilon_t,$$

and test the residual

$$\hat{\varepsilon} = w_{n,t} - \hat{\beta}_0 - \hat{\beta}_1 w_{1,t} - \cdots - \hat{\beta}_{n-1} w_{n-1,t}$$

for a unit root.

Can use the Cointegrating ADF or Cointegrating Regression DW test for this: rejecting H_0 implies $\hat{\varepsilon}_t \sim I(0)$, and therefore, evidence of cointegration.

Properties of OLS for Cointegrated Variables

If y_t and x_t are cointegrated, then OLS estimator $\hat{\beta}_0, \hat{\beta}_1$ in

$$y_t = \beta_0 + \beta_1 x_t + \varepsilon_t$$

will be **superconsistent**.

Normally,

$$\sqrt{T}(\hat{\beta} - \beta) \longrightarrow \mathcal{N}(0, \mathbf{V}).$$

That is, $\hat{\beta}$ converges at rate $\sqrt{T}$.

For cointegrated time-series,

$$T(\hat{\beta} - \beta) \longrightarrow \mathcal{N}(0, \mathbf{V}).$$

That is, $\hat{\beta}$ converges at rate T . This implies that when **cointegrating relationships exist**, OLS estimates **reliably even in small samples**; when they **do not exist**, OLS leads to **spurious relationships**.

Income Consumption Example

Define:

- c_t : (log) consumption;
- y_t : (log) income;
- a_t : (log) wealth;

and arrange them into a 3×1 vector $w_t = (c_t, y_t, a_t)'$.

Extension of the **Permanent Income Hypothesis** states that for $\beta = (1, -1, \beta)'$, $z_t = \beta' w_t = c_t - y_t - \beta a_t$ is stationary.

To test this, regress

$$c_t - y_t = \beta_0 + \beta_1 a_t + \varepsilon_t,$$

and test for stationarity of $\hat{\varepsilon} = c_t - y_t - \hat{\beta}_0 - \hat{\beta}_1 a_t$.

Use, e.g., **Cointegrating ADF** with (in this case) two variables—Table C of Enders.

Cointegration Space

All examples so far assumed cointegrating vectors are either **fully known** or **partially known**.

In the permanent income hypothesis example,

$$\mathbf{B} = \begin{pmatrix} 1 \\ -1 \\ \beta \end{pmatrix}.$$

In the RBC example, we can generalize slightly to obtain

$$\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \beta_1 & \beta_2 \end{pmatrix}.$$

In all these cases, the cointegrating vectors are **unique** and we can estimate unknown components such as β_1 and β_2 from data.

Cointegration Space

In general, if $\mathbf{z}_t = \mathbf{B}'\mathbf{w}_t$ is stationary, then for any invertible $\mathbf{K}$, $\tilde{\mathbf{B}} = \mathbf{BK}$ is another $n \times r$ matrix that yields $\tilde{\mathbf{z}}_t = \tilde{\mathbf{B}}'\mathbf{w}_t = \mathbf{K}'\mathbf{z}_t$ stationary.

Multiplying $\mathbf{B}$ by $\mathbf{K}$ yields **linear combinations** of the vectors in $\mathbf{B}$. The r vectors formed by these linear combinations are also valid cointegrating vectors.

Implication: unrestricted $\mathbf{B}$ **cannot** be estimated from data because only the **space** spanned by (true) cointegrating vectors $\beta_1, \dots, \beta_r$ (termed **cointegration space**) is identified.

For identification, restrictions on $\mathbf{B}$ are necessary:

- Impose 0 and 1 restrictions based on theoretical considerations (as in examples so far), or
- Impose the restriction $\mathbf{B}'\mathbf{B} = \mathbf{I}_r$, i.e., identify the basis of the **cointegration space**.